Name_____ Date_____ Period_____

LESSON 2-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Building a Box Plot

Maryland Standard 6.SP.B.4 · Learning target – I can find the five-number summary of a data set and use it to build a box plot.

REMEMBER

- **Five-number summary**: minimum, Q1, median, Q3, maximum.
- **Q1** is the median of the lower half; **Q3** of the upper half. Odd count: leave the median out.
- The **box** goes from Q1 to Q3, with a line at the median. **Whiskers** reach the min and max.

WORKED EXAMPLE

Find the five-number summary of 4, 6, 9, 10, 13, 15, 18.

Step 1 Min = 4, max = 18, median = 10

Step 2 Lower half 4, 6, 9, so Q1 = 6

Step 3 Upper half 13, 15, 18, so Q3 = 15

Answer **4, 6, 10, 15, 18**

WORD BANK

five-number summary – min, Q1, median, Q3, max

Q1 – middle of the lower half

Q3 – middle of the upper half

box plot – a box from Q1 to Q3 with whiskers

PRACTICE

show your work · label units

1. **Push-ups.** Nine students did 22, 40, 10, 34, 18, 46, 25, 14, and 30 push-ups. Find the five-number summary.

   Step 1 Put the 9 numbers in order.

   Step 2 The median is the 5th number.

   Step 3 Find the middle of each half (leave the median out).

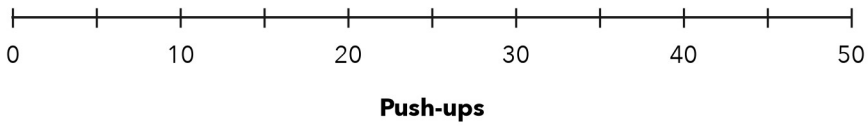
Min	Q1	Median	Q3	Max

2. Draw a box plot of the push-up data from Problem 1.

Step 1 Mark the five numbers above the line.

Step 2 Draw the box from Q1 to Q3 and a line at the median.

Step 3 Draw whiskers to the min and max.



Push-ups

3. **Temperatures.** The box plot shows daily high temperatures (°F) for a month.



Temperature (°F)

a. Median: _____ °F

b. Q1 and Q3: _____

c. Fraction of days from 58 to 70°F: _____

4. **Homework Minutes.** 26, 14, 39, 20, 11, 32, 18, 28. Find each value.

Step 1 Order the 8 numbers.

Step 2 Split them into two halves of 4.

Step 3 Find the middle of each half.

a. Q1: ____ **b.** Median: ____ **c.** Q3: ____

5. **Find the Error.** Grace says Q1 of 2, 5, 7, 8, 12, 14, 20 is 6. She used 2, 5, 7, 8 as the lower half. Explain her error and find Q1.

Frame *Grace should leave ___ out. The lower half is ___, so Q1 = ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Explain to someone at home what the box in a box plot shows.

Name_____ Date_____ Period_____

LESSON 2-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Comparing Box Plots

Maryland Standard 6.SP.B.4 · Learning target – I can compare two groups by reading the medians and box widths of their box plots.

REMEMBER

- Compare **centers**: use the medians.
- Compare **spread**: box width and range.
- A **narrower box** (Q3 − Q1) is more **consistent**.

WORKED EXAMPLE

A: median 18, box 14–22. B: median 24, box 20–26.

Step 1 Center: 24 > 18, so B scored higher.

Step 2 Box widths: A is 22 − 14 = 8; B is 26 − 20 = 6.

Answer **B is higher and more consistent**

WORD BANK

median – the line inside the box

box width – Q3 − Q1

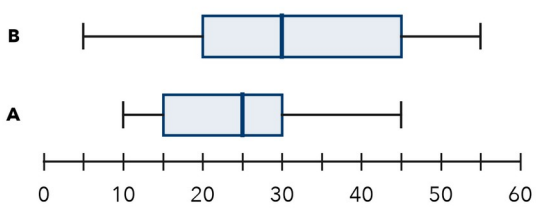
range – max − min

consistent – close together

PRACTICE

show your work · label units

1. Reading Time. Minutes of reading per day for Class A and Class B.



Step 1 Find each median line. Subtract.

Step 2 Box width = Q3 − Q1 for each class.

Step 3 The narrower box is more consistent.

a. Greater median: _____ by _____ min

b. Box widths: A _____ B _____

c. More consistent: _____

2. Find each measure of spread.

a. Q1 = 12, Q3 = 30. Box width: _____

b. Min = 5, max = 41. Range: _____

3. Bowling. Complete the table for two bowlers' scores.

Step 1 Box width = Q3 − Q1.

Step 2 Range = max − min.

	Min	Q1	Median	Q3	Max	Box width	Range
Maya	80	95	102	110	125		
Leon	70	90	105	112	140		

Who is more consistent?_____

4. Plant Heights. Both medians are 14 cm. Box S: 12–16 cm. Box T: 8–20 cm. Which varies more?

□ Group S □ Group T

5. Choose a Bus. Both medians are 20 min. Boxes: A 18–22, B 14–26 min. Which is more reliable?

Frame *Bus ___ is more reliable because its box is ___.*

TEST PRACTICE

choose the best answer

6. X: Q1 = 22, Q3 = 34. Y: Q1 = 25, Q3 = 31. Which middle half is more consistent?

A X **B** Y **C** They are equal **D** Cannot tell

7. A box plot has the summary 8, 12, 15, 21, 30. How wide is the box?

A 9 **B** 22 **C** 15 **D** 3

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___